PG No. 29 $T_{\rm h}$ $m\bar{3}$ [cubic]

* generator : $2_{001},\ 2_{010},\ 3^{+}_{111},\ -1$
* conjugacy class

$[1]:$ 1
$[2_{001}]:$ $2_{001},\ 2_{100},\ 2_{010}$
$[3^{+}_{111}]:$ $3^{+}_{111},\ 3^{+}_{1-1-1},\ 3^{+}_{-11-1},\ 3^{+}_{-1-11},\ 3^{-}_{111},\ 3^{-}_{1-1-1},\ 3^{-}_{-11-1},\ 3^{-}_{-1-11}$
$[-1]:$ -1
$[\mathrm{m}_{001}]:$ $\mathrm{m}_{001},\ \mathrm{m}_{100},\ \mathrm{m}_{010}$
$[-3^{+}_{111}]:$ $-3^{+}_{111},\ -3^{+}_{1-1-1},\ -3^{+}_{-11-1},\ -3^{+}_{-1-11},\ -3^{-}_{111},\ -3^{-}_{1-1-1},\ -3^{-}_{-11-1},\ -3^{-}_{-1-11}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	1	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	1
2	2_{001}	$\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&1\end{bmatrix}$	1
3	2_{010}	$\begin{bmatrix}-1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$	1
4	2_{100}	$\begin{bmatrix}1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}$	1
5	3^{+}_{111}	$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	1
6	3^{+}_{-11-1}	$\begin{bmatrix}0&0&1\\-1&0&0\\0&-1&0\end{bmatrix}$	1
7	3^{+}_{1-1-1}	$\begin{bmatrix}0&0&-1\\-1&0&0\\0&1&0\end{bmatrix}$	1
8	3^{+}_{-1-11}	$\begin{bmatrix}0&0&-1\\1&0&0\\0&-1&0\end{bmatrix}$	1
9	3^{-}_{111}	$\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	1
10	3^{-}_{1-1-1}	$\begin{bmatrix}0&-1&0\\0&0&1\\-1&0&0\end{bmatrix}$	1
11	3^{-}_{-1-11}	$\begin{bmatrix}0&1&0\\0&0&-1\\-1&0&0\end{bmatrix}$	1

continued ...

Table 1

No.	tag	matrix (polar)	det
12	3^{-}_{-11-1}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$	1
13	-1	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
14	m_{001}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
15	m_{010}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1
16	m_{100}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1
17	-3^{+}_{111}	$\begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	-1
18	-3^{+}_{-11-1}	$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	-1
19	-3^{+}_{1-1-1}	$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	-1
20	-3^{+}_{-1-11}	$\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	-1
21	-3^{-}_{111}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}$	-1
22	-3^{-}_{1-1-1}	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$	-1
23	-3^{-}_{-1-11}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$	-1
24	-3^{-}_{-11-1}	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$	-1